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ICE 10.1



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1. (2 marks) Let $q=11$. Which of the following is a "primitive root" of $q=11$? Hint: Look at the table given below which gives $a^x \pmod{q}$ for different values of a, x .

- 2, 3, 4, 5, and 6
- 3, 4, 5, 9, and 10
- **2, 6, 7, and 8 (correct)**
- 2, 3, 4, 5, 6, 7, 8, 9, and 10
- None of the above

2. (2 marks) This is about Alice and Bob computing a shared key using the Diffie Hellman algorithm. Let $q=19, a=2$. If Alice select $X_A=5$, then what is the corresponding public key Alice sends to Bob?

- 6
- **13 (correct)**
- 7
- None of the above

3. (2 marks) Revisit the problem given in 2 above. If Bob selects $X_B=6$, then what is the corresponding public key Bob sends to Alice?

- 17
- 13
- **7 (correct)**
- None of the above

4. (2 marks) In continuation with questions 2 and 3, what is the shared key computed by Alice or Bob?

- **11 (correct)**
- 12
- 15
- None of the above

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